

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase IIIb, Multicenter, Open-Label, Single-Arm Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Subcutaneous Emicizumab in Patients From Birth to 12 Months of Age With Hemophilia A Without Inhibitors
Short Title:	A Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Subcutaneous Emicizumab in Participants From Birth to 12 Months of Age With Hemophilia A Without Inhibitors
Protocol Number:	1114781949771846518
NCT Number:	NCT04431726
Sponsor Name:	Roche
Investigational Product:	emicizumab
Phase:	PHASE3
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. Model-Based Annualized Bleeding Rate for Treated Bleeds: The number of treated bleeds over the efficacy period was estimated as an annualized bleeding rate (ABR) using a negative binomial regression model, which accounts for different follow-up times. A treated bleed was defined as a bleed that was directly followed by a hemophilia medication reported to be a "treatment for bleed". The 72-hour rule was implemented: two bleeds of the same type and at the same anatomical location were counted as one bleed if the second bleed occurred within 72 hours from the last treatment for the first bleed. Bleeds due to surgery/procedure were excluded. 2. Mean Calculated Annualized Bleeding Rate for Treated Bleeds: The number of treated bleeds over the efficacy period is presented here as a calculated annualized bleeding rate (ABR) that was annualized for each participant using the following formula: $ABR = (\text{number of bleeds} / \text{number of days during the efficacy period}) \times 365.25$. A treated bleed was defined as a bleed that was directly followed by a hemophilia medication reported to be a "treatment for bleed". The 72-hour rule was implemented: two bleeds of the same type and at the same anatomical location were counted as one bleed if the second bleed occurred within 72 hours from the last treatment for the first bleed. Bleeds due to surgery/procedure were excluded. 3. Median Calculated Annualized Bleeding Rate for Treated Bleeds: The number of treated bleeds over the efficacy period is presented here as a calculated annualized bleeding rate (ABR) that was annualized for each participant using the following formula: $ABR = (\text{number of bleeds} / \text{number of days during the efficacy period}) \times 365.25$. A treated bleed was defined as a bleed that was directly followed by a hemophilia medication reported to be a "treatment for bleed". The 72-hour rule was
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implemented: two bleeds of the same type and at the same anatomical location were counted as one bleed if the second bleed occurred within 72 hours from the last treatment for the first bleed. Bleeds due to surgery/procedure were excluded.

Secondary Obj.:

1. Hemophilia Joint Health Score (HJHS) Total Score at Specified Timepoints During the Long-Term Follow-Up Period
2. Magnetic Resonance Imaging (MRI) Score of Specific Joints at Specified Timepoints During the Long-Term Follow-Up Period
3. Incidence and Severity of Adverse Events, With Severity Determined According to World Health Organization (WHO) Toxicity Grading Scale

2.2 Background & Rationale

This is a Phase IIIb, multicenter, open-label, single-arm study of prophylactic emicizumab in previously untreated and minimally treated patients at study enrollment from birth to ≥ 12 months of age with severe hemophilia A (intrinsic factor VIII [FVIII] level $<1\%$) without FVIII inhibitors. The study is designed to evaluate the efficacy, safety, pharmacokinetics, and pharmacodynamics of emicizumab administered at 3 milligrams per kilogram of body weight (mg/kg) once every 2 weeks (Q2W) for 52 weeks. After 1 year of treatment, participants will continue to receive emicizumab (1.5 mg/kg once every week [QW], 3 mg/kg Q2W or 6 mg/kg once every 4 weeks [Q4W]) over a 7-year long-term follow-up period under this study frame.

3. Study Design & Methodology

Design Framework: Type: INTERVENTIONAL, Phase: PHASE3, Status: ACTIVE_NOT_RECRUITING

Target N: 55

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

- * Age from birth to ≥ 12 months at time of informed consent
- * Body weight ≥ 3 kilograms (kg) at time of informed consent. Patients with a lower body weight can be enrolled after they have reached a body weight of 3 kg. Premature babies (gestational age <38 weeks) may be enrolled as long as they have reached a body weight of 3 kg. For premature babies, the corrected gestational age should be reported.
- * Mandatory receipt of vitamin K prophylaxis according to local standard practice
- * Diagnosis of severe congenital hemophilia A (intrinsic FVIII level $<1\%$)
- * A negative test for FVIII inhibitor (i.e., <0.6 Bethesda units [BU]/mL) locally assessed during the 2-week screening period
- * No history of documented FVIII inhibitor (i.e., <0.6 BU/mL), FVIII drug-elimination half-life <6 hours, or FVIII recovery $<66\%$
- * Previously untreated patients or minimally treated patients (i.e., up to 5 days of exposure with hemophilia-related treatments, such as plasma-derived FVIII, recombinant FVIII, fresh frozen plasma, cryoprecipitate, or whole blood products)
- * Documentation of the details of the hemophilia-related treatments received since birth
- * Documentation of the details of the bleeding episodes since birth
- * For patients from birth to <3 months of age at the time of study entry: no evidence of active intracranial hemorrhage,

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as confirmed by a negative cranial ultrasound at screening irrespective of delivery mode

* Adequate hematologic, hepatic, and renal function, as defined in the protocol

* For parents/caregivers: willingness and ability to comply with the study protocol requirements, scheduled visits, treatment plans, laboratory tests, completion of applicable questionnaires, and other study procedures

4.2 Exclusion Criteria

* Inherited or acquired bleeding disorder other than severe hemophilia A

* Use of systemic immunomodulators (e.g., interferon) at enrollment or planned use during the study

* Receipt of any of the following: Prior use of emicizumab prophylaxis including investigational or commercial emicizumab; An investigational drug to treat or reduce the risk of hemophilic bleeds within 5 drug-elimination half-lives of last drug administration; A non-hemophilia-related investigational drug within the last 30 days or 5 drug-elimination half-lives, whichever is shorter; An investigational drug concurrently

* Current active severe bleed, such as intracranial hemorrhage

* Planned surgery (excluding minor procedures, e.g., circumcision, CVAD placement) during the study

* History of clinically significant hypersensitivity associated with monoclonal antibody therapies or components of the emicizumab injection

* Patients who are at high risk for thrombotic microangiopathy (TMA) (e.g., have a previous medical or family history of TMA, such as thrombotic thrombocytopenic purpura, atypical hemolytic uremic syndrome) in the investigator's judgment

* Previous or current treatment for thromboembolic disease (with the exception of previous catheter-associated thrombosis in patients for whom anti-thrombotic treatment is not currently ongoing) or signs of thromboembolic disease

* Any hereditary or acquired maternal condition that may predispose the patient to thrombotic events (e.g., inherited thrombophilias antiphospholipid syndrome)

* Other diseases (e.g., certain autoimmune diseases) that may increase risk of bleeding or thrombosis

* Known infection with HIV, hepatitis B virus, or hepatitis C virus

* Serious infection requiring antibiotics or antiviral treatments within 14 days prior to screening

* Concurrent disease, treatment, abnormality in clinical laboratory tests, vital signs measurements, or physical examination findings that could interfere with the conduct of the study or that would, in the opinion of the investigator or Sponsor, preclude the patient's safe participation in and completion of the study or interpretation of the study results

* Unwillingness of the parent or caregiver to allow receipt of blood or blood products, or any standard-of-care treatment for a life-threatening condition

* Any other medical, social, or other condition that may prevent adequate compliance with the study protocol in the opinion of the investigator

11. Study Identifiers & Governance

Primary ID: 1114781949771846518

Registry Number: NCT04431726

Sponsor: Roche

Study Title: A Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Subcutaneous Emicizumab in Participants From Birth to 12 Months of Age With Hemophilia A Without Inhibitors

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15. Sign-off & Approval

Principal Investigator: _____ Date: _____
Clinical Operations Lead: _____ Date: _____
Lead Statistician: _____ Date: _____